

Lecture 6: Investment

Energy Economics of the Green Transition

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Today

1. All the ingredients: the model of *Energy System Models*, section 7
2. The Model Lab and the Model Explorer
3. Part 1: read the reference system
4. Part 2: shock it, with carbon, fuel and capital
5. Part 3: margins of response

While we start: open the Model Lab and make your code,
championape-model-lab.share.connect.posit.cloud.



All the ingredients

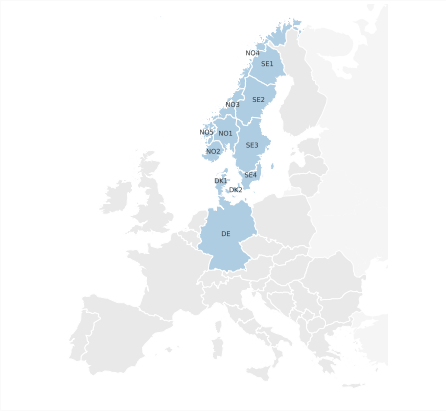
We now have all the ingredients

- **Dispatch** (lecture 3): the merit order; prices as marginal costs and scarcity rents.
- **Wind and solar** (lecture 3): variable output; capture prices and value factors.
- **Storage** (lecture 4): energy moved in time; the arbitrage band.
- **Flexible demand and trade** (lecture 5): demand that responds; zones joined by wires.
- **Investment** (today): capacity becomes a choice, at its annual cost.

With the last ingredient in place, the model answers the transition's own question: what to build, how to run it, and what prices that system makes.

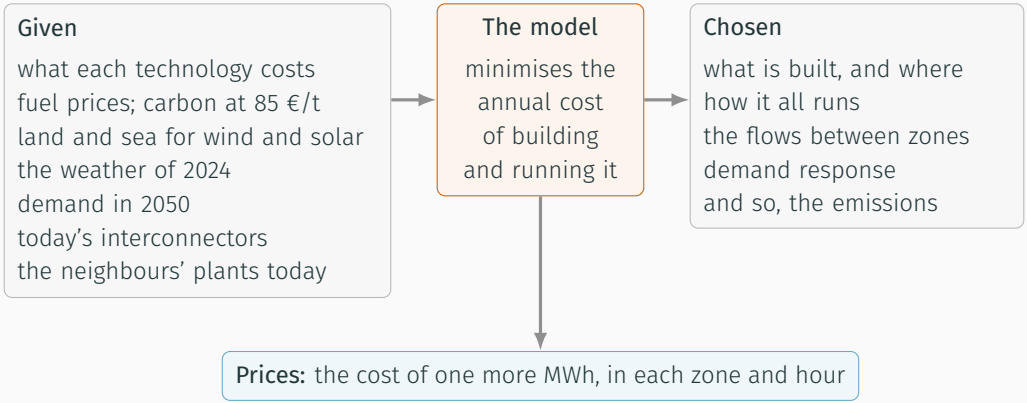
The model: what should Denmark build?

- **The question:** what should Denmark build for a decarbonised power system in 2050, and how should it run it, at the lowest total cost?
- **Twelve zones:** DK1 and DK2, Germany, five in Norway and four in Sweden.
- **Denmark is built from scratch;** the neighbours keep today's plants and may add more.
- **One carbon price,** 85 €/t, everywhere.



Energy System Models, Fig. 6.1: the twelve zones of the model.

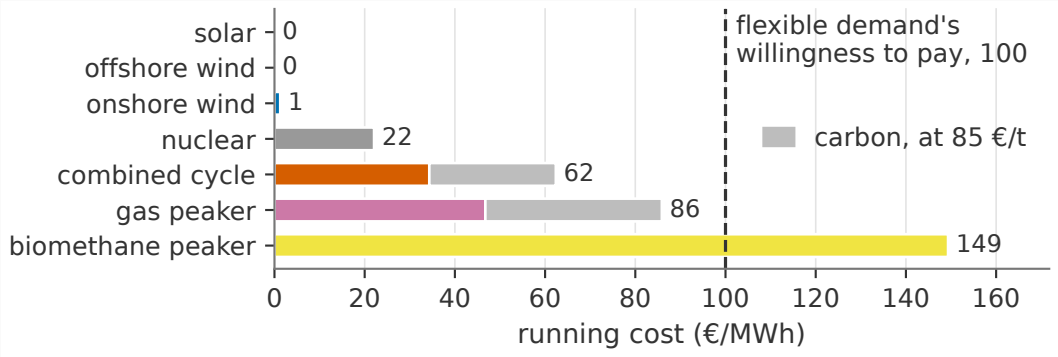
What the model is given, and what it chooses



The Model Lab's description of the model; ESM §7.2–7.3.

What its prices mean

Market prices: a zone's price is the cost of one more MWh there.



Drawn from the Explorer's data (Money › Running costs, and demand's willingness to pay): DK1, reference case. Off the scale: interruptible industry at 3,000 €/MWh, lost load at 23,400.

The Lab and the Explorer

The Model Lab

- **Like our worksheets:** today's questions, part by part; submit a part to see the keys and explanations.
- **With an AI tutor:** feedback on your open answers, which you can revise.
- **And an exercise bank** to study for the exam. If it works, it grows: earlier and later lectures, and exam-style questions with suggested solutions.

Your code is your login: write it down. On paper: the worksheet.



championape-model-lab.
share.connect.posit.cloud

The Model Explorer

- **Lectures 6 and 7:** how to analyse a (relatively) large energy system model.
- **The Explorer** collects its results: the reference case and about a hundred scenarios, each with one given changed.
- **Use it** to analyse how the energy system works, and how it responds when a given changes: compare a scenario with the reference.

championape.github.io/EnergyEconomics-explorer

Part 1

Setup: the greenfield problem

$$\min \quad \sum_h \sum_g (c_g + \tau e_g)\, q_{g,h} + \sum_g \left(a(r, n_g) + f_g\right) I_g\, \bar{q}_g$$

s.t.	$\sum_{g \in \mathcal{G}_z} q_{g,h} + \sum_\ell A_{z\ell} f_{\ell,h} = D_{z,h}$	$(\lambda_{z,h})$	balance
	$0 \le q_{g,h} \le \gamma_{g,h}\, \bar{q}_g$	$(\underline{\mu}_{g,h},\, \overline{\mu}_{g,h})$	availability
	$0 \le \bar{q}_g \le \bar{Q}_g$	$(\underline{\beta}_g,\, \overline{\beta}_g)$	potential

and storage and the network, as in ESM §4 and §6.

ESM eq. (40): generation $q_{g,h}$, availability $\gamma_{g,h}$, capacity $\bar{q}_g$ and its potential $\bar{Q}_g$; the annual fixed cost is the annuity $a(r, n_g)$ plus fixed O&M f_g, times the investment I_g.

Setup: where a generator enters the Lagrangian

The terms of ESM's Lagrangian (41) in which generator g, in zone z, appears:

$$\mathcal{L} = \underbrace{\sum_h \lambda_{z,h}\, q_{g,h}}_{\text{balance}} - \underbrace{\sum_h (c_g + \tau e_g)\, q_{g,h}}_{\text{running cost}} - \underbrace{\big(a(r, n_g) + f_g\big) I_g\, \bar{q}_g}_{\text{fixed cost}}$$

$$+ \underbrace{\sum_h \underline{\mu}_{g,h}\, q_{g,h} + \sum_h \overline{\mu}_{g,h}\big(\gamma_{g,h}\bar{q}_g - q_{g,h}\big)}_{\text{availability}} + \underbrace{\underline{\beta}_g\, \bar{q}_g + \overline{\beta}_g\big(\bar{Q}_g - \bar{q}_g\big)}_{\text{potential}} + \dots$$

Read the reference system

1. Optimal investment: derive Result 6; where $\overline{\beta}_g > 0$; capture price against levelised cost.
2. The generation mix: solar against onshore wind; a peaker's condition; what drives the gas turbines.
3. What sets the price: the margin, month by month; why Danish plants set it so rarely; the highest prices.

In the Lab: lecture 6, part 1, submitted as a whole. The stretch, optimal battery investment, is for those who finish early.

Solutions 1

Solutions 1, Q1–Q2: the questions

Q1. Derive Result 6. For which technologies is $\overline{\beta}_g > 0$? How are capture price and levelised cost related?

Q2. Solar against onshore wind; a peaker's condition on prices; what drives the gas turbines.

Question

Onshore wind in DK2 earns 145 % of its annual fixed cost. Why does the model not build more of it?

1. more would push its capture price below its cost
2. its sites are used up: the potential binds
3. the model limits what a technology may earn
4. wind is too variable to build more of

Solutions 1, Q1: Result 6, rents against fixed cost

$$\overline{\beta}_g - \underline{\beta}_g = \underbrace{\sum_h \gamma_{g,h} \max\left(\lambda_h - c_g - \tau e_g,\ 0\right)}_{\text{a MW's scarcity rent}} - \underbrace{\left(a(r, n_g) + f_g\right) l_g}_{\text{its annual fixed cost}}$$

Built, below its potential

$\overline{\beta}_g = \underline{\beta}_g = 0$

a MW earns exactly its fixed cost

At its potential

$\overline{\beta}_g > 0$

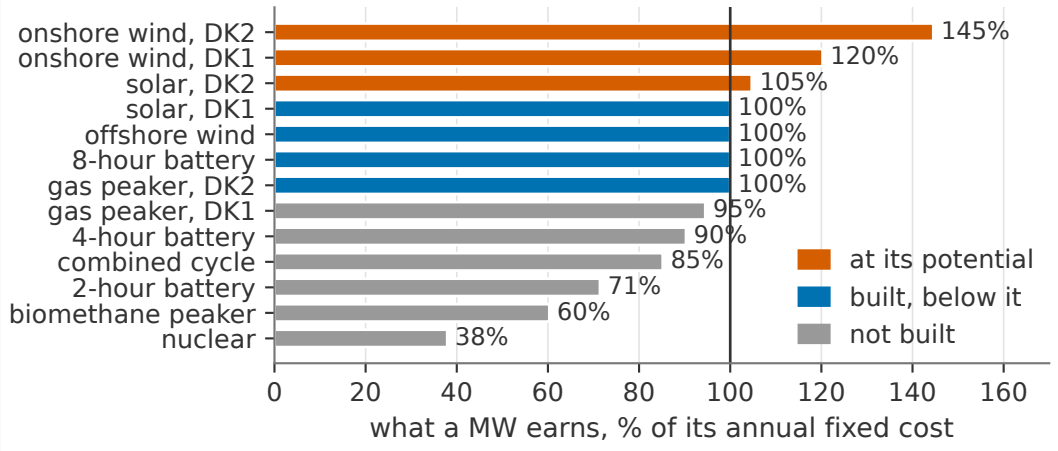
a land rent on the scarce sites

Not built

$\underline{\beta}_g > 0$

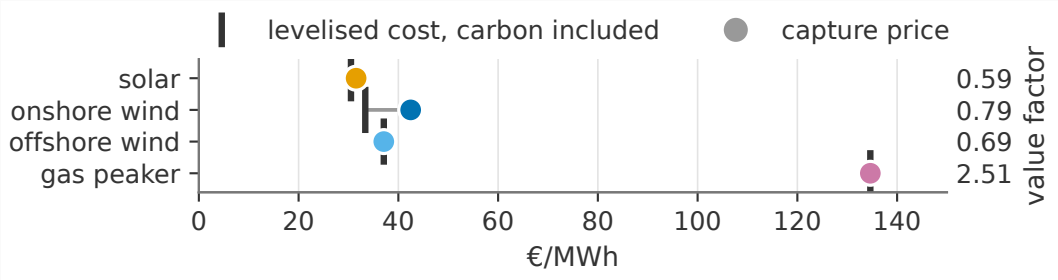
a MW would not pay for itself

Solutions 1, Q1: the three cases in the reference



Drawn from the Explorer's data (Money › Rent per MW against annual fixed cost); ESM Fig. 7.2(b), by zone. Not built: the better of the two zones.

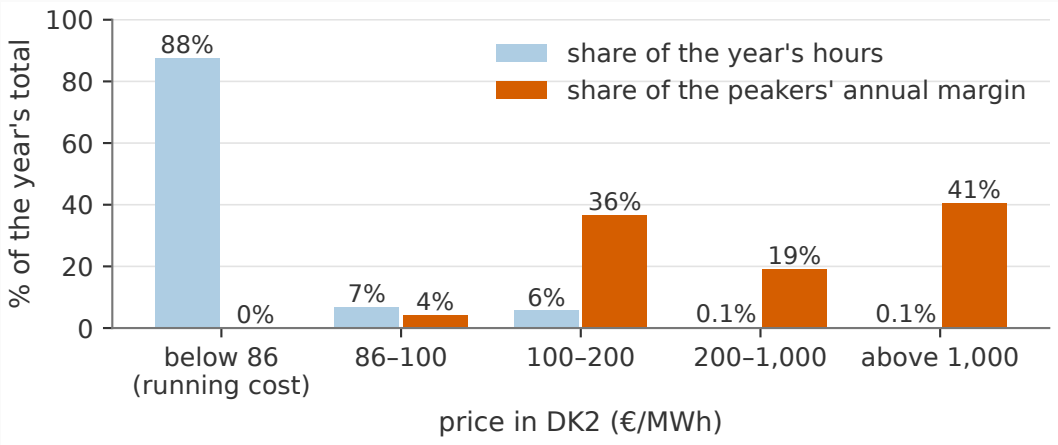
Solutions 1, Q1–Q2: capture price against levelised cost



Drawn from the Explorer's data (Money › Levelised cost and capture price); ESM Table 5. Each levelised cost is at the hours its technology runs in the solution.

- **Q1. Built below its potential,** a technology captures its levelised cost; at its potential, more: the land rent.
- **Q2. Solar is cheaper per MWh, yet onshore wind earns the rent:** its output sells in better hours.

Solutions 1, Q2: a peaker sells availability

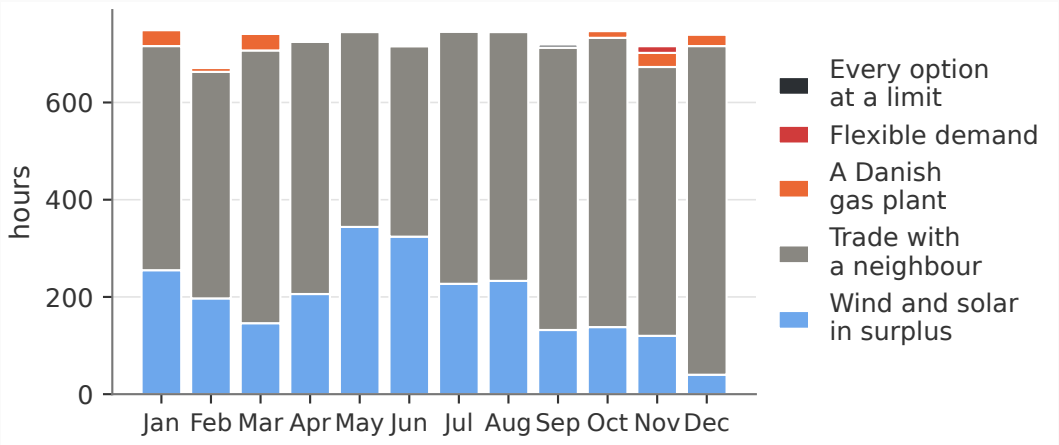


Drawn from the Explorer's data (Money › Peakers' margin by price band); ESM Fig. 7.3. The DK2 price, reference case.

Solutions 1, Q3: the questions

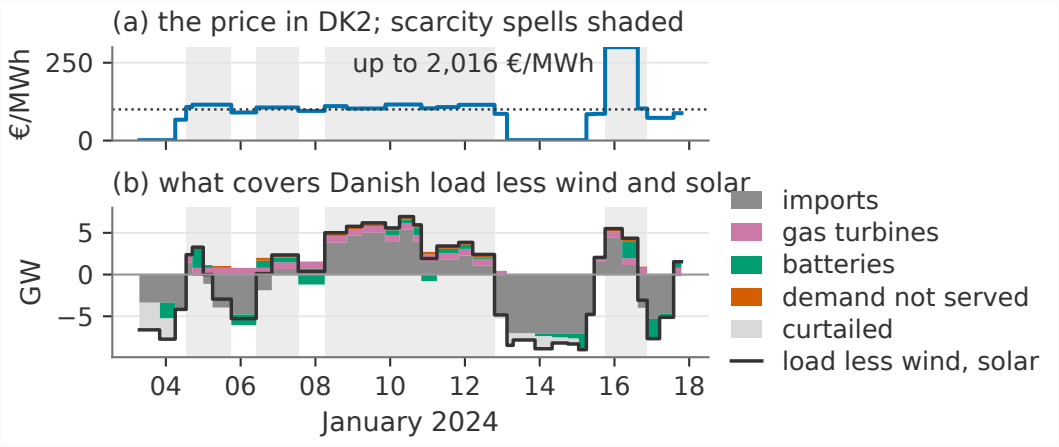
Q3. What is at the margin of the DK2 price, month by month; why Danish plants and consumers set it so rarely; what is at the margin in the hours with the highest prices, and what decides how high the price goes.

Solutions 1, Q3: the Danish price is mostly made abroad



Drawn from the Explorer's data, as Operation › What is at the margin, month by month draws it: DK2, reference case.

Solutions 1, Q3: the highest prices ration what is left



ESM Fig. 7.4 (a) and (b), drawn from the note's results; dotted: flexible demand's willingness to pay. The peak hour itself: Operation › One hour, inspected, for DK2.

Part 2

Setup: one lever at a time

- **Each scenario** changes one given and solves the whole model again.
- **Predict first,** then compare the scenario with the reference in the Explorer.

Part 2 needs nothing from part 1.

Shock it: carbon, fuel and capital

1. A higher carbon price: predict.
2. Denmark's marginal abatement cost curve.
3. Expensive gas: predict; why no combined cycle; why more peakers; what takes gas's place.
4. Cheap capital: predict; why the discount rate matters more for wind and solar.

In the Lab: lecture 6, part 2.

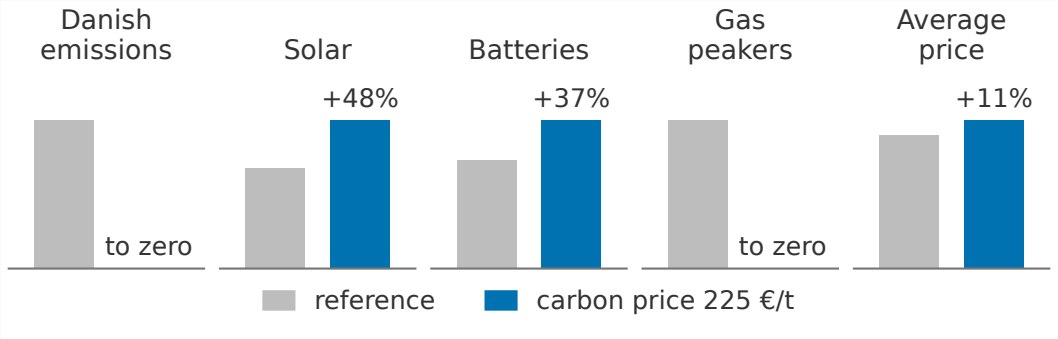
Solutions 2

Solutions 2, Q1–Q2: the questions

Q1. The carbon price rises from 85 to 225 €/t: Danish emissions, solar, batteries, gas peakers, the average price.

Q2. The shape of Denmark's marginal abatement cost curve.

Solutions 2, Q1: a higher carbon price



The Explorer's headline numbers, Carbon price 225 EUR/t against the reference; capacities in Denmark.

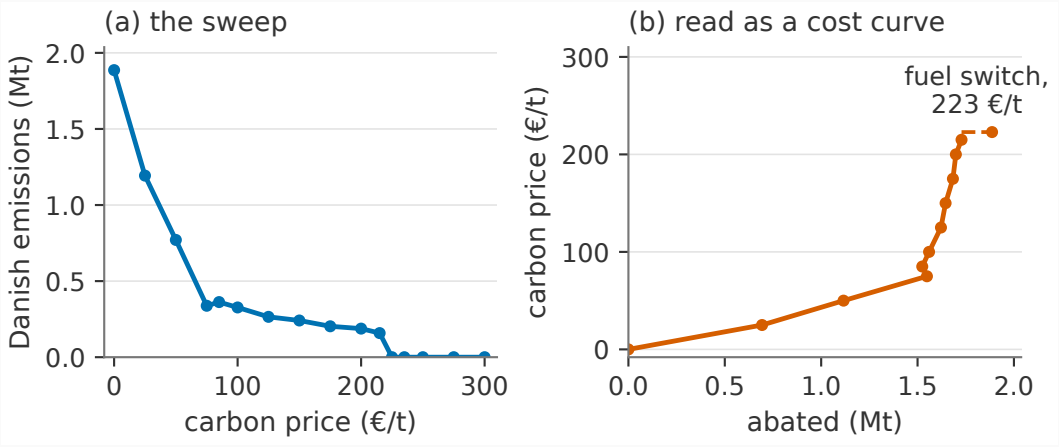
- **Emissions to zero:** at 225 €/t the peaker runs on biomethane, so gas peakers go.
- **The price rises by only a tenth:** carbon reaches it only in the hours an emitter is at the margin.

Question

Danish emissions fall to zero in one step near 220 €/t. What sets that price?

1. the cost of building biomethane turbines
2. the extra fuel cost of biomethane, per tonne of CO_2 avoided
3. the price at which Danish wind and solar sites run out
4. the price at which an 8-hour battery pays for itself

Solutions 2, Q2: Denmark's marginal abatement cost curve



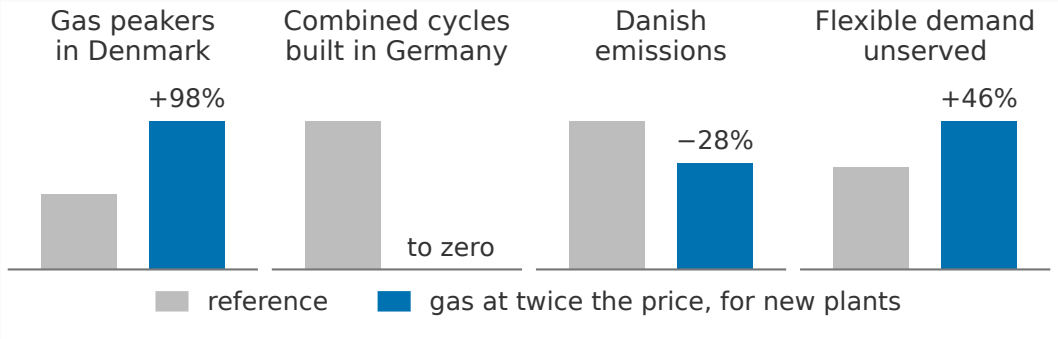
Drawn from the Explorer's data: (a) Sweeps, Danish emissions against Carbon price, every zone; (b) the same scenarios with the axes swapped.

Solutions 2, Q3–Q4: the questions

Q3. Gas doubles for new plants: predict; why Germany stops building combined cycles while Denmark builds more peakers; what takes gas's place.

Q4. Capital at 2 %: predict; why the discount rate matters more for wind and solar than for a gas plant.

Solutions 2, Q3: expensive gas



The Explorer's headline numbers, Gas at 2 x, new plants only, against the reference. ESM Fig. 7.6 shows it beside the other shocks.

Solutions 2, Q3: a gas plant has two jobs

Energy, over thousands of hours

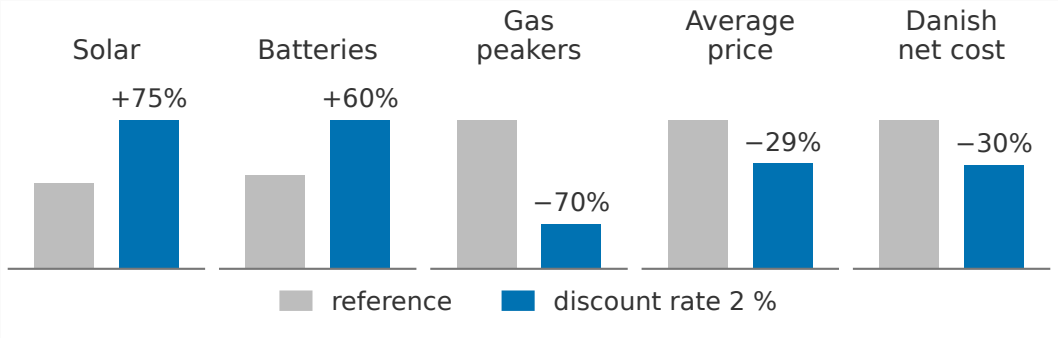
Once gas is expensive, wind, solar and batteries do it more cheaply: Germany builds no combined cycle.

Standing by, in winter spells

It came free with the combined cycles. Without them scarce hours come more often: Denmark doubles its peakers.

What takes gas's place: German solar and batteries for its energy; Danish demand response for its last hours.

Solutions 2, Q4: a system that is almost all capital



The Explorer's headline numbers, Discount rate 2 % against the reference; capacities in Denmark.

- **Wind, solar and batteries** cost almost only their annuity, which rises with *r*; a gas plant's cost is mostly fuel.
- **The cost of capital is the price level:** 96 % of the Danish bill is capital.

Part 3

Setup: margins of response

- **A system adapts to a shock through its margins of response:** building something else, storing, trading, or demand response.
- **Part 3 takes them away, one at a time,** to compare with the reference.

Margins of response

1. No batteries: predict; why demand response is called on less.
2. Batteries and peakers: predict 2-hour batteries; a summer and a winter week; two kinds of shortfall.
3. Demand response: predict; what the model says about Brown et al. (2025).
4. Denmark's borders: autarky; a Danish budget; why leakage shrinks.

In the Lab: lecture 6, part 3.

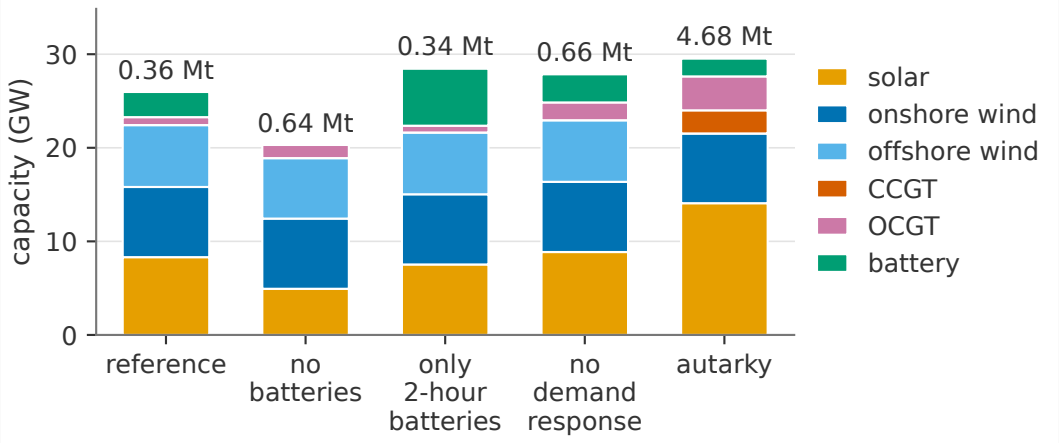
Solutions 3

Solutions 3, Q1–Q4: the questions

Q1. No batteries. **Q2.** Only 2-hour batteries.

Q3. No demand response. **Q4.** Autarky, and a Danish budget.

Solutions 3: margins of response are substitutes



Drawn from the Explorer's data: Danish capacity, with Danish emissions above the bars. Take a margin away, and the others carry the adjustment, at a cost.

Wrap-up

What carries forward

1. **Prices recover capital.** What is built earns exactly its fixed cost; a scarce site earns a rent (Result 6).
2. **A renewable system's price is made by trade, surplus and scarcity.** The few expensive hours pay for the capacity that stands by.
3. **Margins of response substitute for one another,** at a cost; a territorial target leaks across a border.

Lecture 7, Monday 26 October: the whole neighbourhood under one carbon budget. Read *Energy System Models*, sections 8 and 9, and appendix D.

Model answers to today's open questions: in the Lab tomorrow. The written solutions follow.

References i

Brown, Tom, Fabian Neumann, and Iegor Riepin (2025). **“Price Formation Without Fuel Costs: The Interaction of Demand Elasticity with Storage Bidding”.** *Energy Economics* 147, p. 108483.